

Math 118– Lecture 01

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1 Main idea

Write the one-sentence point of today's lecture here. Basic information about everything.

Chapter 0 is preliminaries, inner product spaces

2 Definitions and notation

Definition 2.1 (Previous Notes). Recall in $\mathbb{C}$,

$$z = re^{i\theta} = r(\cos \theta + i \sin \theta)$$

Notice this makes multiplication pretty nice in $\mathbb{C}$, as angles combine easily

Definition 2.2 (Power Series of e (Taylor Series)). For $z \in \mathbb{C}$, e^z is defined by

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$
$$\sum_{n=0}^{\infty} \frac{(z+w)^n}{n!} = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\sum_{k=0}^n \binom{n}{k} z^k w^{n-k} \right)$$

The $n!$ cancel out and then you can rearrange into (NUMBERING THE LATTICE)

$$\sum_{k=0}^{\infty} \sum_{n=k}^{\infty} \frac{1}{k!(n-k)!} z^k w^{n-k}$$

Rearrange

$$j = n - k \rightarrow \sum_{k=0}^{\infty} \sum_{j=0}^{\infty} \frac{1}{k!} z^k \frac{1}{j!} w^j = \left(\sum_{k=0}^{\infty} \frac{1}{k!} z^k \right) \left(\sum_{j=0}^{\infty} \frac{1}{j!} w^j \right)$$

Definition 2.3 (Derivation of De Moivre's in a different way). If you look at

$$e^{i\theta} = \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!}$$

which is

$$\left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots\right) + i \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots\right) = \cos \theta + i \sin \theta$$

Definition 2.4 (Remember These Properties of Conjugation).

$$\overline{\overline{z}w} = \overline{z} \cdot \overline{w}, \quad \overline{\left(\frac{z}{w}\right)} = \frac{\overline{z}}{\overline{w}}$$

If

$$f(z) = \sum_{n=0}^{\infty} c_n (z - a)^n, c_n a \in \mathbb{R}$$

then

$$f(\overline{z}) = \overline{f(z)}$$

Definition 2.5 (Vector Space). Vector space over a field like real or complex where you can perform addition and multiply by scalars, etc...

$\mathbb{C}^n$ is basically a vector space over $\mathbb{R}$ of dimension $2n$. (but only real operations)

Infinite Dimensional Spaces (like the set of all polynomials)

For example, $l^2(\mathbb{N})$ = space of infinite sequences $(x_1, x_2, \dots)$ such that the sum $\sum_{n=1}^{\infty} |x_n|^2$ converges.

For example,

$$l^1(\mathbb{Z}) = \text{doubly infinite, summable sequences } \sum_{x=1}^{\infty} |x_i| < \infty, \text{ Show that } l^1(\mathbb{N}) \text{ is a subset of } l^2(\mathbb{N})$$

Proof: If you look at some $x \in l^1(\mathbb{N})$, we have $\sum_{n=1}^{\infty} |x_i|$

You can just look at the expansion of $C^2 = \sum_{a=1}^{\infty} |x_a| \sum_{b=1}^{\infty} |x_b|$ which is just the terms we want plus a lot of extraneous absolute value terms bounding $\sum_{i=1}^{\infty} |x_i^2|$ converges.

For example,

$$L'([a, b]) = f : [a, b] \rightarrow \mathbb{C} \text{ such that } \int_a^b |f(x)| dx \text{ converges}$$

One can basically show $L^2([a, b]) \subseteq L'([a, b])$ like $f(x) = \frac{1}{\sqrt{x}}$ which is in $L'([0, 1])$ but not $L^2([0, 1])$

For example, $L^4(\mathbb{R}^2)$ is a function $\mathbb{R}^2 \rightarrow \mathbb{C}$ such that

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |f(x, y)|^4 dx dy \text{ converges}$$

3 Claims, theorems, and proof sketches

Claim. State the claim in one precise sentence.

Proof sketch. Record the key idea, the first useful equation, and the step that still needs checking.

4 Examples and calculations

Example 4.1 (Lecture example). Write the setup, the calculation, and the conclusion.

5 Questions and follow-up

- What is still unclear?
- Which exercise or reading should I revisit?
- TODO: